

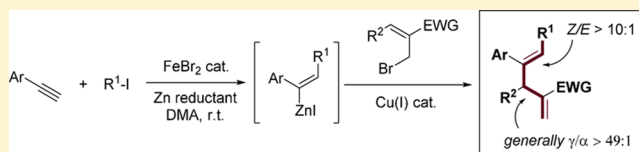
γ -Selective Allylation of (*E*)-Alkenylzinc Iodides Prepared by Reductive Coupling of Arylacetylenes with Alkyl Iodides

Fedor E. Zhurkin and Xile Hu*

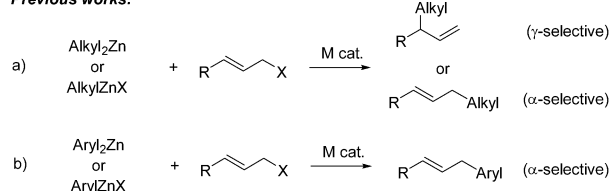
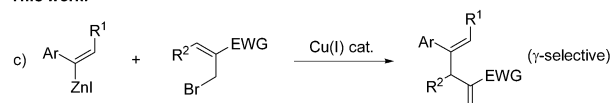
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S Supporting Information

ABSTRACT: The first examples of Cu-catalyzed γ -selective allylic alkenylation using organozinc reagents are reported. (*E*)-Alkenylzinc iodides were prepared by Fe-catalyzed reductive coupling of terminal arylalkynes with alkyl iodides. In the presence of a copper catalyst, these reagents reacted with allylic bromides derived from Morita–Baylis–Hillman alcohols to give 1,4-dienes in high yields. The reactions are highly γ -selective (generally $\gamma/\alpha > 49:1$) and tolerate a wide range of functional groups such as ester, cyano, keto, and nitro.

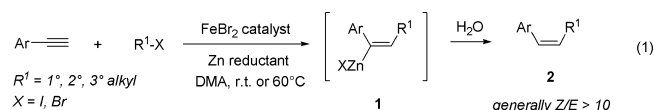


Metal-catalyzed allylic substitution using organozinc reagents is a versatile method for C–C bond formation. Unlike alkylzinc^{1–3} and arylzinc^{4,5} reagents (Scheme 1, a and b) which are widely used in these reactions, alkenylzinc reagents are rarely employed. There are only a few examples of catalytic allylic alkenylation using alkenylzinc reagents to yield 1,4-dienes, which are ubiquitous in nature and represent an important class of synthetic building blocks.^{5–10}

Scheme 1. Metal-Catalyzed Allylic Substitution with Organozinc Reagents**Previous works:****This work:**

Alkenylzinc compounds are commonly prepared¹¹ by direct Zn insertion into the carbon–halogen bond of alkenyl halides,^{12,13} or by transmetalation from alkenyl organometallic reagents.¹⁴ Both approaches are limited by the difficulty to obtain stereochemically pure alkenyl halides.¹⁵ Alkenylzinc reagents can also be prepared by carbozincation of alkynes with organozinc reagents.¹⁶ This approach is potentially stereoselective, but it requires reactive organometallic reagents which can be hard to handle or can lower functional group compatibility. We have recently reported Fe-catalyzed reductive coupling of arylacetylenes with alkyl halides to form *cis*-alkenes. Mechanistic studies indicated that the reaction proceeded via

formation of (*E*)-alkenylzinc intermediates (eq 1).^{17,18} Thus, this reaction provides an easy access to stereochemically pure



alkenylzinc reagents **1** without the need for sensitive organometallic reagents. Here we describe a Cu-based catalytic system for the reactions of these alkenylzinc reagents with allylic bromides (Scheme 1c). To the best of our knowledge, this is the first γ -selective catalytic allylic alkenylation. The few precedents of catalytic allylic alkenylation either employed symmetrical allylic substrates^{12,19–21} or were α -selective.^{22,23}

Allylic halides derived from Morita–Baylis–Hillman (MBH) alcohols were chosen because the products contain both an activated C–C double bond and an electron-withdrawing group that are prone to further transformations. A diverse number of substrates are available thanks to the large scope of MBH reactions.²⁴ Moreover, earlier works^{25–27} showed that alkylation of MBH alcohols-derived allylic halides could be γ -selective. It was found that under Cu-catalysis ethyl (*Z*)-2-(bromomethyl)-3-phenyl acrylate (**3a**) reacted with **1a**, prepared by Fe-catalyzed reductive coupling of phenylacetylene and iodocyclohexane, to give the corresponding γ -product **4a** with high regioselectivity (Table 1). Among various copper catalysts, CuCN·2LiCl gave the highest yield and γ -selectivity (Table 1, entries 1–6). The yield could be improved using a longer reaction time (24 h) and a slower addition rate with 5 mol % of CuCN·2LiCl as the catalyst (Table 1, entry 7). Replacing dichloromethane (DCM) by tetrahydrofuran (THF) as the solvent led to a lower yield (Table 1, entry 8). The highest yield was obtained using 5 mol % of CuCN·2LiCl in DCM at $-30\text{ }^{\circ}\text{C}$ for 24 h with 0.25 mmol

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Table 1. Optimization of Reaction Conditions of (*E*)-Alkenylzinc Iodides Allylation

entry	addition rate (mL/h)	catalyst (mol %)	solvent (mL)	time (h)	yield ^b (%) (4a:4ab)
1 ^c	3.0	CuBr·SMe ₂ (10)	DCM (2)	3.5	58 (71:1)
2 ^c	3.0	(MeCN) ₄ CuPF ₆ (10)	DCM (2)	3.5	63 (66:1)
3 ^c	3.0	(CuOTf) ₂ ·C ₆ H ₆ (10)	DCM (2)	3.5	53 (40:1)
4 ^c	3.0	CuCl·2LiCl (10)	DCM (2)	3.5	65 (45:1)
5 ^c	3.0	CuCN·2LiCl (10)	DCM (2)	3.5	74 (70:1)
6 ^c	3.0	CuNHC ^d (5)	DCM (2)	3.5	<1 (n.d.)
7 ^c	1.0	CuCN·2LiCl (5)	DCM (2)	24	88 (56:1)
8 ^c	1.0	CuCN·2LiCl (5)	THF (2)	24	77 (42:1)
9 ^e	1.0	CuCN·2LiCl (5)	DCM (2)	24	93 ^f (68:1)
10 ^e	1.0	—	DCM (2)	24	1 (n.d.)

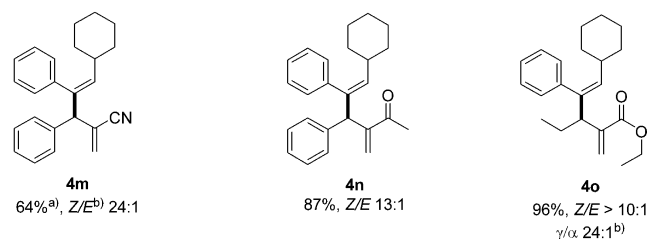
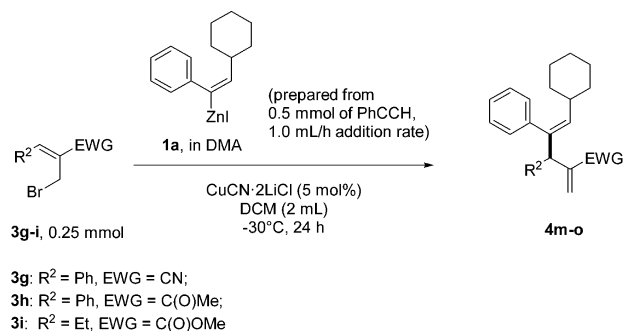
^aPhenylacetylene (0.5 mmol, 1 equiv), iodocyclohexane (1.5 equiv), Zn (1.5 equiv), TMSCl (20 mol %), and FeBr₂ (10 mol %) were stirred in DMA (1 mL) overnight (17–19 h). The resulting solution was used directly for the reactions with allylic bromides. ^bUncalibrated GC yield of the isomeric mixture (4a + 4aa + 4ab), corrected by number of carbons. ^c0.35 mmol of 3a was used. ^dChloro[1,3-bis(2,6-diisopropylphenyl)imidazol-2-ylidene]copper(I) was used as catalyst. ^e0.25 mmol of 3a was used. ^f4a:4aa 12:1.

of 3a and 1a prepared on 0.5 mmol scale (Table 1, entry 9). Only a trace amount of product was obtained without a copper catalyst (Table 1, entry 10).

Table 2 shows the scope of this reaction with respect to (*E*)-alkenylzinc reagents. Despite the presence of a complex mixture of metal ions and unreacted organic starting materials and side products, the desired dienes 4a–g were obtained in high isolated yields and with high *Z/E* selectivity. In all cases the products were almost exclusively γ -regioisomers ($\gamma/\alpha > 49:1$). Various substituents on the aryl rings of the (*E*)-alkenylzinc reagents were tolerated (1a–1d, 1g). The R¹ group of the (*E*)-alkenylzinc reagents can be either cyclic (1a, 1f) or acyclic (1d, 1e).

The scope of 2-(bromomethyl)acrylates is shown in Table 3. Both aryl- (3a–e) and heteroaryl-substituted (3f) substrates could be used. The reactions were again highly γ -selective

Scheme 2. Scope of Novel Types of MBH Alcohols-Derived Allylic Bromides



^aIsolated yield of pure product (as a mixture of isomers). ^bDetermined by ¹H NMR (similar values were obtained by GC).

($\gamma/\alpha > 49:1$) and *Z*-selective. Electron-withdrawing substituents in the aryl ring generally led to high yields, except for 4k, which has a nitro group in *ortho*-position of the phenyl ring. The steric influence of this nitro group probably led to the modest yield (49%). An electron-rich 2-furylsubstituted substrate also reacted in a high yield (3f). A crystal structure of compound 4k was determined (Figure 1) to confirm the regio- and diastereoselectivity of the reaction.

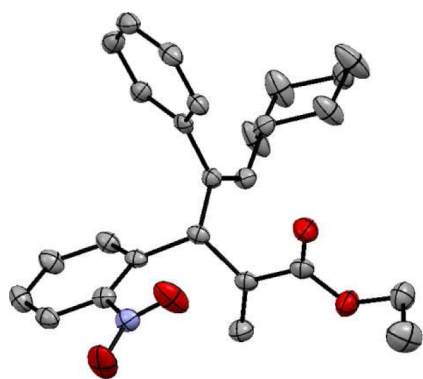


Figure 1. X-ray structure of compound 4k. Thermal ellipsoids are drawn at 50% probability level. Hydrogen atoms are omitted. Color code: red for O; blue for N; gray for C.

Table 2. Scope of (*E*)-Alkenylzinc Reagent

entry	alkenylzinc iodide	product	yield ^a , %	Z/E ^b
1			87	12:1
2			84	12:1
3			79	11:1
4			73	57:1
5			74	21:1
6			68	16:1
7			73	11:1

^aIsolated yields of pure products (as a mixture of isomers). ^bDetermined by ¹H NMR. Similar values were obtained by GC. ^cExo/endo of the corresponding alkene is >50:1.¹⁷

Previously allylic substitution with MBH alcohols-derived allylic bromides was limited to reactions of 3-aryl-2-(bromomethyl)-acrylates.²⁵ In the current study, the scope of allylic bromides is increased (Scheme 2). Substrates bearing a nitrile group (3g) and alkyl keto group (3h) reacted with excellent regioselectivity ($\gamma/\alpha > 49:1$). A substrate derived from alkyl aldehyde (3i) was also alkenylated with good regioselectivity ($\gamma/\alpha = 24:1$).

In summary, the first Cu-catalyzed γ -selective allylic alkenylation was developed, employing (*E*)-alkenylzinc reagents prepared

by Fe-catalyzed reductive coupling of arylacetylenes with alkyl iodides and allylic bromides derived from Morita–Baylis–Hillman alcohols. The method uses a simple copper(I) catalyst and tolerates a number of important functional groups such as ester, nitrile, keto, and nitro. This method provides an easy access to highly functionalized 1,4-dienes in high regio- and Z/E-selectivity and may be used to prepare libraries of steroid mimics and antitumor drugs, such as aromatase inhibitor tamoxifen and related compounds.²⁸

Table 3. Scope of Aryl- and Heteroaryl-Substituted 2-(Bromomethyl)acrylates

entry	allylic bromide	product	yield ^a , %	Z/E ^b
1			87	12:1
2			90	12:1
3			84	12:1
4			92	13:1
5			49	16:1
6			75	11:1

^aIsolated yield of pure product (as a mixture of isomers). ^bDetermined by ¹H NMR (similar values were obtained by GC).

EXPERIMENTAL SECTION

General Information. NMR spectra were recorded on a 400 MHz instrument at ambient temperature in CDCl₃ as solvent. ¹H NMR chemical shifts (δ, ppm) were measured relative to the tetramethylsilane (TMS) signal in CDCl₃ (0.00 ppm) unless otherwise stated. Splitting patterns are designated as s, singlet; d, doublet; t, triplet; q, quartet; m, multiplet; br, broad. ¹³C NMR chemical shifts (δ, ppm) are reported relative to the CDCl₃ signal (77.16 ppm) unless otherwise stated. The diffraction data were measured at low temperature [100(2) K] using Mo K_α radiation on a diffractometer equipped with a kappa geometry goniometer.

Unless otherwise noted, all chemicals were commercially available and were used as received without further purifications. Solvents were purified using a two-column solid-state purification system and transferred to glovebox without exposure to air by the aid of a Straus flask. Zn powder (<10 μ, 98%+) was purchased from Aldrich. Anhydrous dimethylacetamide (DMA) (99.8% purity) was commercially purchased

and stored under nitrogen. Iron(II) bromide (FeBr₂, 98% purity) was purchased from Aldrich or Acros. All the chiral starting materials and products were in the form of racemic mixtures; for the products containing two stereogenic centers the corresponding diastereomeric ratio was 1:1. Silica gel (40–63 μm, 230–400 mesh) was used as the stationary phase for column chromatography.

Starting Materials Preparation. For additional details about the preparation of (*E*)-alkenylzinc reagents **1a–g**, see refs 17 and 18. All the substrates **3a–i** were prepared from corresponding Morita–Baylis–Hillman (MBH) alcohols by treatment with HBr/H₂SO₄²⁵ or PBr₃.²⁹ For the furyl-substituted substrate **3f**, PBr₃ must be used. In order to obtain good yields in allylic substitution reactions, the crude bromides should be purified by column chromatography or (if possible) recrystallized from ether/hexane.

For the substrates **3a–e**, **3g**, the starting MBH-alcohols can be prepared by using the procedure from ref 25. MBH-alcohols for

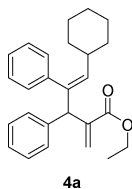
preparation of **3f** and **3i** are synthesized using a 1,4-dioxane/water 1:1 mixture as solvent, in order to accelerate the reaction.³⁰ In the case of the alcohol that corresponds to the bromide **3h**,³¹ we were unable to separate it from the methylvinylketone (MVK) dimer that forms as a side product. However, after treatment of the product mixture by PBr₃, the resulting bromide **3h** can be easily isolated from the unreacted MVK dimer. All the bromides are obtained as *Z*-isomers,^{25,31} with the exception of **3g**, which was obtained as a mixture of *E*- and *Z*-isomers.³²

General Procedures. Alkenylzinc Reagent Solution. Under a dry nitrogen atmosphere a 20 mL screw-cap vial, equipped with a magnetic stirring bar, was charged with Zn dust (49 mg, 0.75 mmol), DMA (1 mL), and TMSCl (11 mg, 0.1 mmol). The mixture was vigorously shaken for a while, and then FeBr₂ (11 mg, 0.05 mmol), alkyne (0.5 mmol), and alkyl iodide (0.75 mmol) were added. The vial was then sealed, and its content was allowed to stir overnight (18–20 h) at ambient temperature.

Allylic Alkenylation. A 20 mL screw-cap vial, equipped with a small magnetic stirring bar (to prevent splashing the reaction mixture on the walls of the vial), was charged under a dry nitrogen atmosphere with allylic bromide (0.25 mmol, 1.0 equiv), CuCN·2LiCl (0.13 mL of 0.1 M solution in THF, 0.013 mmol, 5 mol %), and dry degassed DCM (2 mL). The vial was sealed with a rubber septum, taped, and cooled to –30 °C. Alkenylzinc reagent solution was added using a syringe pump at 1.0 mL/h rate. The resulting mixture was allowed to stir for 24 h at –30 °C (counting from the beginning of organozinc reagent addition). The mixture was quenched with 1 M HCl. *n*-Dodecane (57 µL, 0.25 mmol) was added, and the mixture was extracted into ca. 4 mL of diethyl ether. The ethereal layer was analyzed by GC-MS at this point. Afterward, the content of the vial was poured into water or 1 M HCl and extracted with diethyl ether (4 × 10 mL). Combined ether extracts were dried over Na₂SO₄ and concentrated *in vacuo*. The crude was dried *in vacuo* and purified by column chromatography (10–15 g of SiO₂, ethyl acetate/hexane 1:99–10:90).

If the GC yield is less than 80–85%, to facilitate the chromatographic isolation of the product, the crude can be stirred with DABCO (0.25–0.5 mmol) in 2–4 mL of diethyl ether overnight. Then, the ether was removed *in vacuo* and the residue was taken up in a small amount of the eluent for the column chromatography purification.

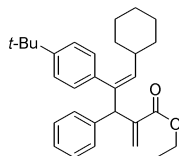
Ethyl (Z)-5-Cyclohexyl-2-methylene-3,4-diphenylpent-4-enoate (4a). Prepared from phenylacetylene (51 mg, 0.5 mmol),



4a

iodocyclohexane (158 mg, 0.75 mmol), and bromide **3a** (67 mg, 0.25 mmol). Yellowish oil, 80.9 mg (87%). *Z/E* 12:1. ¹H NMR (CDCl₃, 400 MHz): 7.29–7.13 (10H, m); 6.28 (1H, s); 5.24 (1H, d, *J* = 9.9 Hz); 5.08 (1H, s); 4.91 (1H, s); 4.23 (2H, q, *J* = 7.0 Hz); 2.1–2.0 (1H, m); 1.64–1.49 (5H, m); 1.30 (3H, t, *J* = 7.0 Hz); 1.17–1.03 (5H, m). ¹³C NMR (CDCl₃, 101 MHz): 167.5; 144.1; 142.2; 139.8; 139.0; 136.2; 129.7; 128.6; 128.4; 128.0; 126.8; 126.6; 60.9; 55.0; 37.7; 33.6; 33.3; 26.1; 25.7; 25.7; 14.5. Anal. Calcd for C₂₆H₃₀O₂: C, 83.38; H, 8.07. Found: C, 83.47; H 8.18. HRMS (ESI-TOF) *m/z*: [M + Na]⁺ Calcd for C₂₆H₃₀O₂Na 397.2143; Found 397.2144.

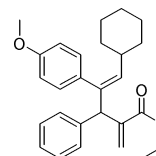
Ethyl (Z)-4-(4-(*tert*-Butyl)phenyl)-5-cyclohexyl-2-methylene-3-phenylpent-4-enoate (4b). Prepared from 1-(*tert*-butyl)-4-ethynylbenzene (79 mg, 0.5 mmol), iodocyclohexane (158 mg, 0.75 mmol), and bromide **3a** (67 mg, 0.25 mmol). Yellowish oil, 90.3 mg (84%).



4b

Z/E 12:1. ¹H NMR (CDCl₃, 400 MHz): 7.29–7.15 (7H, m); 7.06 (2H, d, *J* = 7.8 Hz); 6.27 (1H, s); 5.21 (1H, d, *J* = 10.0 Hz); 5.04 (1H, s); 4.92 (1H, s); 4.21 (2H, q, *J* = 7.0 Hz); 2.1–2.0 (1H, m); 1.68–1.50 (5H, m); 1.31–1.27 (12H, m); 1.16–1.06 (5H, m). ¹³C NMR (CDCl₃, 101 MHz): 167.5; 149.1; 144.4; 140.2; 139.0; 138.5; 136.4; 129.7; 128.3; 128.1; 126.7; 126.4; 124.8; 60.9; 54.8; 37.6; 34.5; 33.7; 33.4; 31.5; 26.1; 25.7; 25.7; 14.5. Anal. Calcd for C₃₀H₃₈O₂: C, 83.67; H 8.89. Found: C, 83.67; H 9.13. HRMS (ESI-TOF) *m/z*: [M + H]⁺ Calcd for C₃₀H₃₈O₂ 431.2945; Found 431.2946.

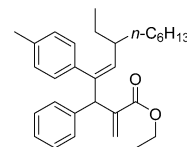
Ethyl (Z)-5-Cyclohexyl-4-(4-methoxyphenyl)-2-methylene-3-phenylpent-4-enoate (4c). Prepared from 1-ethynyl-4-methoxybenzene



4c

(66 mg, 0.5 mmol), iodocyclohexane (158 mg, 0.75 mmol), and bromide **3a** (67 mg, 0.25 mmol). Yellowish oil, 80.2 mg (79%). *Z/E* 11:1. ¹H NMR (CDCl₃, 400 MHz): 7.28–7.16 (5H, m); 7.06 (2H, d, *J* = 7.9 Hz); 6.76 (2H, d, *J* = 7.9 Hz); 6.27 (1H, s); 5.21 (1H, d, *J* = 9.9 Hz); 5.05 (1H, s); 4.88 (1H, s); 4.22 (2H, q, *J* = 7.1 Hz); 3.73 (3H, s); 2.1–2.0 (1H, m); 1.64–1.43 (5H, m); 1.30 (3H, t, *J* = 7.1 Hz); 1.19–1.02 (5H, m). ¹³C NMR (CDCl₃, 101 MHz): 167.5; 158.3; 144.2; 139.9; 138.5; 136.0; 134.5; 129.7; 129.6; 128.4; 126.7; 126.4; 113.4; 60.9; 55.2; 55.1; 37.7; 33.7; 33.3; 26.1; 25.8; 25.7; 14.5. Anal. Calcd for C₂₇H₃₂O₃: C, 80.16; H, 7.97. Found: C, 80.06; H, 8.16.

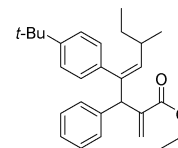
Ethyl (Z)-6-Ethyl-2-methylene-3-phenyl-4-(*p*-tolyl)dodec-4-enoate (4d). Prepared from 1-ethynyl-4-methylbenzene (58 mg,



4d

0.5 mmol), 3-iodononane (191 mg, 0.75 mmol), and bromide **3a** (67 mg, 0.25 mmol). Yellowish oil, 78.5 mg (73%). *Z/E* 57:1. ¹H NMR (CDCl₃, 400 MHz): 7.29–7.18 (5H + 5H, m); 7.01 (4H + 4H, br.s.); 6.30 (1H + 1H, s); 5.12 (1H + 1H, br.s.); 5.10 (1H + 1H, d, *J* = 10.7 Hz); 4.93–4.92 (1H + 1H, m); 4.29–4.16 (2H + 2H, m); 2.26 (3H + 3H, s); 2.01 (1H + 1H, br.s.); 1.34–1.05 (15H + 15H, m); 0.90–0.84 (6H + 3H, m), 0.72 (3H, t, *J* = 7.4 Hz). ¹³C NMR (CDCl₃, 101 MHz): 167.53; 167.48; 144.02; 140.98; 140.91; 139.93; 139.46; 139.41; 135.82; 135.21; 135.16; 129.85; 129.82; 128.74; 128.72; 128.58; 128.35; 126.76; 126.74; 126.62; 126.59; 60.97; 60.96; 55.56; 40.00; 39.87; 36.06; 35.82; 32.10; 32.06; 29.74; 29.72; 29.09; 28.85; 27.71; 27.25; 22.91; 22.82; 21.26; 14.40; 14.29; 14.26; 12.33; 11.85. Anal. Calcd for C₃₀H₄₀O₂: C, 83.28; H, 9.32. Found: C, 83.53; H 9.40. HRMS (ESI-TOF) *m/z*: [M + Na]⁺ Calcd for C₃₀H₄₀O₂Na 455.2926; Found 455.2922.

Ethyl (Z)-4-(4-(*tert*-Butyl)phenyl)-6-methyl-2-methylene-3-phenyloct-4-enoate (4e). Prepared from 1-(*tert*-butyl)-4-ethynylbenzene

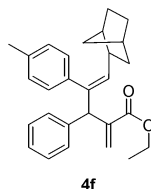


4e

(79 mg, 0.5 mmol), 2-iodobutane (138 mg, 0.75 mmol), and bromide **3a** (67 mg, 0.25 mmol). Yellowish oil, 74.4 mg (74%). *Z/E* 21:1. ¹H NMR (CDCl₃, 400 MHz): 7.29–7.19 (7H + 7H, m); 7.07–7.03 (2H + 2H, m); 6.30–6.28 (1H + 1H, m); 5.18–5.10 (2H + 1H, m); 5.03 (1H, m); 4.93 (1H, s); 4.91 (1H, s); 4.28–4.18 (2H + 2H, m); 2.21–2.10 (1H + 1H, m); 1.34–1.26 (14H + 14H, m); 0.98 (3H, d,

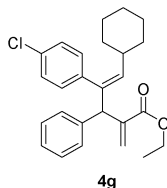
$J = 6.6$ Hz); 0.85–0.80 (3H + 3H, m); 0.71 (3H, t, $J = 7.4$ Hz). ^{13}C NMR (CDCl_3 , 101 MHz): 167.51; 167.49; 149.07; 149.04; 144.44; 144.06; 140.17; 140.02; 139.64; 139.48; 139.15; 139.13; 136.71; 136.29; 129.77; 129.76; 128.34; 128.19; 128.15; 126.71; 126.70; 126.57; 126.53; 124.80; 124.77; 60.96; 60.88; 55.07; 54.97; 34.80; 34.51; 34.50; 31.49; 30.45; 30.41; 21.66; 21.14; 14.45; 14.42; 12.31; 11.96. Anal. Calcd for $\text{C}_{28}\text{H}_{36}\text{O}_2$: C, 83.12; H, 8.97. Found: C, 83.17; H 9.06. HRMS (ESI-TOF) m/z : $[\text{M} + \text{Na}]^+$ Calcd for $\text{C}_{28}\text{H}_{36}\text{O}_2\text{Na}$ 427.2613; Found 427.2614.

Ethyl (Z)-5-(Bicyclo[2.2.1]heptan-2-yl)-2-methylene-3-phenyl-4-(p-tolyl)pent-4-enoate (4f). Prepared from 1-ethynyl-4-methylbenzene



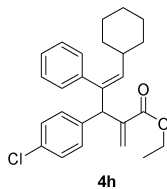
(58 mg, 0.5 mmol), 2-iodobicyclo[2.2.1]heptane (167 mg, 0.75 mmol), and bromide **3a** (67 mg, 0.25 mmol). Yellowish oil, 67.9 mg (68%). *Z/E* 16:1. ^1H NMR (CDCl_3 , 400 MHz): 7.29–7.15 (5H + 5H, m); 7.07–7.02 (4H + 4H, m); 6.27 (1H + 1H, br.s.); 5.27–5.23 (1H + 1H, m); 5.09 (1H, s); 5.06 (1H, s); 4.95–4.93 (1H + 1H, m); 4.25–4.18 (2H + 2H, m); 2.28 (3H + 3H, s); 2.20–2.08 (2H + 2H, m); 2.02 (1H, br.s.); 1.87 (1H, br.s.); 1.48–1.00 (11H + 11H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 167.49; 167.48; 144.24; 144.21; 140.14; 140.12; 139.02; 138.97; 137.89; 137.84; 136.03; 135.95; 129.72; 129.67; 128.77; 128.74; 128.64; 128.40; 128.36; 126.75; 126.71; 126.50; 126.48; 60.87; 54.74; 54.69; 43.55; 43.38; 41.29; 41.24; 40.01; 39.64; 36.76; 36.74; 36.31; 29.58; 28.94; 28.90; 21.29; 14.45; 14.43. Anal. Calcd for $\text{C}_{28}\text{H}_{32}\text{O}_2$: C, 83.96; H, 8.05. Found: C, 83.91; H, 8.18.

Ethyl (Z)-4-(4-Chlorophenyl)-5-cyclohexyl-2-methylene-3-phenylpent-4-enoate (4g). Prepared from 1-chloro-4-ethynylbenzene (68 mg,



0.5 mmol), iodocyclohexane (158 mg, 0.75 mmol), and bromide **3a** (67 mg, 0.25 mmol). Yellowish oil, 74.5 mg (73%). *Z/E* 11:1. ^1H NMR (CDCl_3 , 400 MHz): 7.29–7.24 (2H, m); 7.22–7.18 (5H, m); 7.08–7.04 (2H, m); 6.29 (1H, m); 5.27 (1H, d, $J = 9.9$ Hz); 5.06 (1H, m); 4.83 (1H, s); 4.27–4.19 (2H, m); 2.0–1.9 (1H, m); 1.65–1.46 (5H, m); 1.31 (3H, t, $J = 7.1$ Hz); 1.19–1.01 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 167.3; 143.8; 140.6; 139.3; 138.0; 136.5; 132.4; 129.9; 129.7; 128.5; 128.3; 127.0; 126.7; 61.0; 54.9; 37.8; 33.5; 33.1; 26.0; 25.7; 25.6; 14.5. HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd for $\text{C}_{26}\text{H}_{30}\text{ClO}_2$ 409.1929; Found 409.1924.

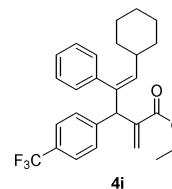
Ethyl (Z)-3-(4-Chlorophenyl)-5-cyclohexyl-2-methylene-4-phenylpent-4-enoate (4h). Prepared from phenylacetylene (51 mg, 0.5 mmol),



iodocyclohexane (158 mg, 0.75 mmol), and bromide **3b** (76 mg, 0.25 mmol). Yellowish oil, 92.1 mg (90%). *Z/E* 12:1. ^1H NMR (CDCl_3 , 400 MHz): 7.25–7.11 (9H, m); 6.30 (1H, s); 5.22 (1H, d, $J = 9.9$ Hz); 5.09 (1H, s); 4.89 (1H, s); 4.23 (2H, q, $J = 7.1$ Hz); 2.1–2.0 (1H, m); 1.65–1.42 (5H, m); 1.30 (3H, t, $J = 7.1$ Hz); 1.19–0.98 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 167.2; 143.7; 141.8; 138.8; 138.4;

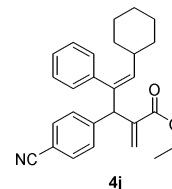
136.5; 132.6; 131.0; 128.6; 128.5; 128.1; 126.8; 126.7; 61.0; 54.4; 37.7; 33.6; 33.2; 26.1; 25.7; 25.7; 14.4. Anal. Calcd for $\text{C}_{26}\text{H}_{29}\text{ClO}_2$: C, 76.36; H, 7.15. Found: C, 76.46; H, 7.28.

Ethyl (Z)-5-Cyclohexyl-2-methylene-4-phenyl-3-(4-(trifluoromethyl)phenyl)pent-4-enoate (4i). Prepared from phenylacetylene



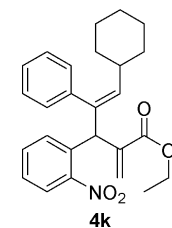
(51 mg, 0.5 mmol), iodocyclohexane (158 mg, 0.75 mmol), and bromide **3c** (84 mg, 0.25 mmol). White solid, 92.8 mg (84%). *Z/E* 12:1. ^1H NMR (CDCl_3 , 400 MHz): 7.54 (2H, d, $J = 7.8$ Hz); 7.34 (2H, d, $J = 7.9$ Hz); 7.26–7.13 (5H, m); 6.33 (1H, s); 5.25 (1H, d, $J = 10.0$ Hz); 5.08 (1H, s); 5.00 (1H, s); 4.23 (2H, q, $J = 7.0$ Hz); 2.1–2.0 (1H, m); 1.66–1.48 (5H, m); 1.31 (3H, t, $J = 7.1$ Hz); 1.20–1.00 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 167.1; 144.2; 143.3; 141.7; 138.4; 137.1; 130.0; 129.2 (q, $J_{\text{C-F}} = 32.4$ Hz); 128.5; 128.2; 126.9; 126.9; 125.4 (q, $J_{\text{C-F}} = 3.8$ Hz); 124.4 (q, $J_{\text{C-F}} = 272.0$ Hz); 61.1; 54.7; 37.7; 33.5; 33.2; 26.1; 25.7; 25.6; 14.4. Anal. Calcd for $\text{C}_{27}\text{H}_{29}\text{F}_3\text{O}_2$: C, 73.28; H, 6.61. Found: C, 73.18; H, 6.73.

Ethyl (Z)-3-(4-Cyanophenyl)-5-cyclohexyl-2-methylene-4-phenylpent-4-enoate (4j). Prepared from phenylacetylene (51 mg, 0.5 mmol),



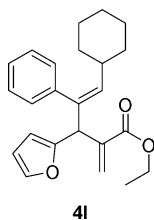
iodocyclohexane (158 mg, 0.75 mmol), and bromide **3d** (74 mg, 0.25 mmol). Yellowish oil, 91.7 mg (92%). *Z/E* 13:1. ^1H NMR (CDCl_3 , 400 MHz): 7.58 (2H, d, $J = 8.2$ Hz); 7.34 (2H, d, $J = 8.2$ Hz); 7.27–7.11 (5H, m); 6.35 (1H, s); 5.22 (1H, d, $J = 10.0$ Hz); 5.10 (1H, s); 5.00 (1H, s); 4.23 (2H, qd, $J = 7.1$ Hz, 2.2 Hz); 2.1–2.0 (1H, m); 1.65–1.48 (5H, m); 1.30 (3H, t, $J = 7.1$ Hz); 1.18–1.01 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 166.8; 145.8; 142.7; 141.3; 138.0; 137.4; 132.3; 130.4; 128.4; 128.2; 127.0; 126.9; 118.9; 110.8; 61.2; 54.9; 37.7; 33.4; 33.1; 26.0; 25.6; 25.6; 14.4. Anal. Calcd for $\text{C}_{27}\text{H}_{29}\text{NO}_2$: C, 81.17; H, 7.32; N, 3.51. Found: C, 81.16; H, 7.39; N, 3.45.

Ethyl (Z)-5-Cyclohexyl-2-methylene-3-(2-nitrophenyl)-4-phenylpent-4-enoate (4k). Prepared from phenylacetylene (51 mg, 0.5 mmol),



iodocyclohexane (158 mg, 0.75 mmol), and bromide **3e** (79 mg, 0.25 mmol). Yellowish solid, 51.9 mg (49%). *Z/E* 16:1. ^1H NMR (CDCl_3 , 400 MHz): 7.85 (1H, d, $J = 8.0$ Hz); 7.58–7.53 (2H, m); 7.39–7.35 (1H, m); 7.28–7.18 (5H, m); 6.42 (1H, s); 5.71 (1H, s); 5.29 (1H, s); 5.13 (1H, d, $J = 10.0$ Hz); 4.19 (2H, q, $J = 7.0$ Hz); 2.1–2.0 (1H, m); 1.62–1.49 (5H, m); 1.25 (3H, t, $J = 7.1$ Hz); 1.15–0.97 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 166.5; 150.0; 141.7; 141.0; 138.2; 137.6; 134.8; 132.5; 131.2; 128.7; 128.2; 127.8; 127.3; 127.0; 125.1; 61.2; 49.6; 37.8; 33.4; 33.2; 26.0; 25.6; 14.3. HRMS (ESI-TOF) m/z : $[\text{M} + \text{Na}]^+$ Calcd for $\text{C}_{26}\text{H}_{29}\text{NNaO}_4$ 442.1994; Found 442.1994.

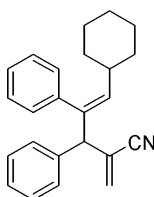
Ethyl (Z)-5-Cyclohexyl-3-(furan-2-yl)-2-methylene-4-phenylpent-4-enoate (4l). Prepared from phenylacetylene (51 mg, 0.5 mmol), iodocyclohexane (158 mg, 0.75 mmol), and bromide **3f** (65 mg,



4l

0.25 mmol). Yellowish oil, 68.6 mg (75%). *Z/E* 11:1. ^1H NMR (CDCl_3 , 400 MHz): 7.35–7.21 (6H, m); 6.33 (1H, br.s.); 6.28 (1H, br.s.); 6.13 (1H, m); 5.27 (1H, br.s.); 5.25 (1H, d, $J = 10.3$ Hz); 4.95 (1H, s); 4.22 (2H, q, $J = 7.1$ Hz); 2.1–2.0 (1H, m); 1.64–1.48 (5H, m); 1.28 (3H, t, $J = 7.1$ Hz); 1.11–0.99 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 166.8; 154.3; 142.1; 141.5; 141.2; 136.9; 136.4; 128.7; 128.0; 126.8; 126.4; 110.2; 108.8; 60.9; 48.6; 37.6; 33.4; 33.2; 26.1; 25.7; 25.6; 14.4. HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd for $\text{C}_{24}\text{H}_{29}\text{O}_3$ 365.2117; Found 365.2119.

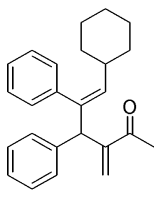
(*Z*)-5-Cyclohexyl-2-methylene-3,4-diphenylpent-4-en-1-one (4m). Prepared from phenylacetylene (51 mg, 0.5 mmol), iodo-



4m

cyclohexane (158 mg, 0.75 mmol), and bromide 3g (56 mg, 0.25 mmol). Yellowish oil, 52.8 mg (64%). *Z/E* 24:1. ^1H NMR (CDCl_3 , 400 MHz): 7.34–7.20 (8H, m); 7.08–7.06 (2H, m); 6.00 (1H, d, $J = 1.1$ Hz); 5.41 (1H, d, $J = 1.5$ Hz); 5.39 (1H, d, $J = 10.1$ Hz); 4.48 (1H, s); 2.1–2.0 (1H, m); 1.69–1.54 (5H, m); 1.21–1.11 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 140.9; 138.8; 137.5; 136.9; 132.3; 129.4; 128.8; 128.5; 128.3; 127.7; 127.0; 126.1; 118.9; 57.7; 38.0; 33.3; 33.1; 26.1; 25.7; 25.6. Anal. Calcd for $\text{C}_{24}\text{H}_{25}\text{N}$: C, 88.03; H, 7.70; N, 4.28. Found: C, 88.09; H, 7.49; N, 4.29.

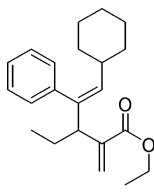
(*Z*)-6-Cyclohexyl-3-methylene-4,5-diphenylhex-5-en-2-one (4n). Prepared from phenylacetylene (51 mg, 0.5 mmol), iodo-



4n

(158 mg, 0.75 mmol), and bromide 3h (60 mg, 0.25 mmol). Yellowish oil, 74.8 mg (87%). *Z/E* 13:1. ^1H NMR (CDCl_3 , 400 MHz): 7.28–7.15 (10H, m); 6.14 (1H, s); 5.35 (1H, d, $J = 1.3$ Hz); 5.12 (1H, d, $J = 9.9$ Hz); 5.04 (1H, s); 2.34 (3H, s); 2.05–1.95 (1H, m); 1.62–1.50 (5H, m); 1.11–0.98 (5H, m). ^{13}C NMR (CDCl_3 , 101 MHz): 199.3; 151.9; 142.1; 140.2; 139.2; 136.3; 129.6; 128.5; 128.4; 128.0; 126.7; 126.6; 53.5; 37.6; 33.6; 33.4; 26.8; 26.1; 25.7; 25.6. HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd for $\text{C}_{25}\text{H}_{29}\text{O}$ 345.2213; Found 345.2204.

Ethyl (*Z*)-5-Cyclohexyl-3-ethyl-2-methylene-4-phenylpent-4-enoate (4o). Prepared from phenylacetylene (51 mg, 0.5 mmol),



4o

iodocyclohexane (158 mg, 0.75 mmol), and bromide 3i (55 mg, 0.25 mmol). Yellowish oil, 78.5 mg (96%). *Z/E* > 10:1. γ/α 24:1. ^1H NMR (CDCl_3 , 400 MHz): 7.30–7.25 (2H, m); 7.23–7.19 (1H, m); 7.09–7.07 (2H, m); 6.20 (1H, d, $J = 0.7$ Hz); 5.34 (1H, m); 5.27 (1H, d, $J = 9.9$ Hz); 4.19 (2H, q, $J = 7.1$ Hz); 3.41 (1H, t, $J = 7.3$ Hz); 1.9–1.8 (1H, m); 1.62–1.47 (7H, m); 1.29 (3H, t, $J = 7.1$ Hz); 1.11–1.02 (5H, m); 0.90 (3H, t, $J = 7.4$ Hz). ^{13}C NMR (CDCl_3 , 101 MHz): 167.9; 142.7; 141.1; 139.3; 135.3; 129.2; 127.8; 126.4; 123.9; 60.7; 50.1; 37.7; 33.5; 33.5; 26.1; 25.7; 25.1; 14.4; 12.5. HRMS (ESI-TOF) m/z : $[\text{M} + \text{H}]^+$ Calcd for $\text{C}_{22}\text{H}_{31}\text{O}_2$ 327.2319; Found 327.2313.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.joc.6b01306.

Copies of NMR spectra and table of X-ray data for 4k (PDF)

Crystallographic data for 4k (CIF)

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Notes

The authors declare no competing financial interest.

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